MSG No. 94.133 $P_C4_22_12$ [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.22`

No.	position	mapping
1	$[0, 0, 0]$	`[1,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,4,5]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,14,15,16]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[10,11,12,13]`

Table 2: Wyckoff site: `4b`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,6,15,16]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,12,13]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,5,10,11]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[7,8,9,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,8,14,15]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,4,11,13]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,5,10,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[6,7,9,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,8,14,15]`
2	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[2,4,11,13]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[3,5,10,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,9,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,6,12,13]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[4,5,9,14]`
4	$\left[0, 0, \frac{3}{4}\right]$	`[7,8,10,11]`

Table 6: Wyckoff site: `4f`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,12,13]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,8,10,11]`

Table 7: Wyckoff site: `8g`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5]`
4	$[0,\ 0,\ -z]$	`[7,8]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[9,14]`
6	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[12,13]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[15,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[4,5]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8]`
5	$\left[\frac{1}{2},\ 0,\ z\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[12,13]`
8	$\left[0,\ \frac{1}{2},\ -z\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,14]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[2,11]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[3,10]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[4,13]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[5,12]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[6,9]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[7,16]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[8,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}]$	`[2,5]`
3	$[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[3,4]`
4	$[-x,\ -x,\ 0]$	`[6,8]`
5	$[x+\frac{1}{2},\ x+\frac{1}{2},\ 0]$	`[9,15]`
6	$[-x,\ x,\ \frac{1}{2}]$	`[10,13]`
7	$[x,\ -x,\ \frac{1}{2}]$	`[11,12]`
8	$[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0]$	`[14,16]`

Table 11: Wyckoff site: `8k`, site symmetry: `..2'`

No.	position	mapping
1	$[x,\ x+\frac{1}{2},\ \frac{1}{2}]$	`[1,15]`
2	$[-x,\ x+\frac{1}{2},\ 0]$	`[2,13]`
3	$[x,\ \frac{1}{2}-x,\ 0]$	`[3,12]`
4	$[x+\frac{1}{2},\ -x,\ 0]$	`[4,11]`
5	$[\frac{1}{2}-x,\ x,\ 0]$	`[5,10]`
6	$[-x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[6,16]`
7	$[x+\frac{1}{2},\ x,\ \frac{1}{2}]$	`[7,9]`
8	$[\frac{1}{2}-x,\ -x,\ \frac{1}{2}]$	`[8,14]`

Table 12: Wyckoff site: `8l`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ \frac{1}{2}]$	`[1,7]`
2	$[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0]$	`[2,5]`
3	$[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0]$	`[3,4]`
4	$[-x,\ -x,\ \frac{1}{2}]$	`[6,8]`
5	$[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	`[9,15]`
6	$[-x,\ x,\ 0]$	`[10,13]`
7	$[x,\ -x,\ 0]$	`[11,12]`
8	$[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[14,16]`

Table 13: Wyckoff site: `8m`, site symmetry: `..2'`

No.	position	mapping
1	$[x,\ x+\frac{1}{2},\ 0]$	`[1,15]`
2	$[-x,\ x+\frac{1}{2},\ \frac{1}{2}]$	`[2,13]`
3	$[x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[3,12]`
4	$[x+\frac{1}{2},\ -x,\ \frac{1}{2}]$	`[4,11]`
5	$[\frac{1}{2}-x,\ x,\ \frac{1}{2}]$	`[5,10]`
6	$[-x,\ \frac{1}{2}-x,\ 0]$	`[6,16]`
7	$[x+\frac{1}{2},\ x,\ 0]$	`[7,9]`

continued ...

Table 13

No.	position	mapping
8	$\left[\frac{1}{2}-x,\,-x,\,0\right]$	[8,14]

Table 14: Wyckoff site: 8n, site symmetry: .2'.

No.	position	mapping
1	$\left[0,\,y,\,\frac{1}{4}\right]$	[1,13]
2	$\left[\frac{1}{2}-y,\,\frac{1}{2},\,\frac{3}{4}\right]$	[2,16]
3	$\left[y+\frac{1}{2},\,\frac{1}{2},\,\frac{3}{4}\right]$	[3,15]
4	$\left[\frac{1}{2},\,\frac{1}{2}-y,\,\frac{1}{4}\right]$	[4,14]
5	$\left[\frac{1}{2},\,y+\frac{1}{2},\,\frac{1}{4}\right]$	[5,9]
6	$\left[0,\,-y,\,\frac{1}{4}\right]$	[6,12]
7	$\left[y,\,0,\,\frac{3}{4}\right]$	[7,11]
8	$\left[-y,\,0,\,\frac{3}{4}\right]$	[8,10]

Table 15: Wyckoff site: 8o, site symmetry: .2'.

No.	position	mapping
1	$\left[0,\,y,\,\frac{3}{4}\right]$	[1,13]
2	$\left[\frac{1}{2}-y,\,\frac{1}{2},\,\frac{1}{4}\right]$	[2,16]
3	$\left[y+\frac{1}{2},\,\frac{1}{2},\,\frac{1}{4}\right]$	[3,15]
4	$\left[\frac{1}{2},\,\frac{1}{2}-y,\,\frac{3}{4}\right]$	[4,14]
5	$\left[\frac{1}{2},\,y+\frac{1}{2},\,\frac{3}{4}\right]$	[5,9]
6	$\left[0,\,-y,\,\frac{3}{4}\right]$	[6,12]
7	$\left[y,\,0,\,\frac{1}{4}\right]$	[7,11]
8	$\left[-y,\,0,\,\frac{1}{4}\right]$	[8,10]

Table 16: Wyckoff site: 16p, site symmetry: 1

No.	position	mapping
1	$[x,\,y,\,z]$	[1]
2	$\left[\frac{1}{2}-y,\,x+\frac{1}{2},\,z+\frac{1}{2}\right]$	[2]
3	$\left[y+\frac{1}{2},\,\frac{1}{2}-x,\,z+\frac{1}{2}\right]$	[3]
4	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,\frac{1}{2}-z\right]$	[4]
5	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,\frac{1}{2}-z\right]$	[5]
6	$[-x,\,-y,\,z]$	[6]
7	$[y,\,x,\,-z]$	[7]
8	$[-y,\,-x,\,-z]$	[8]
9	$\left[x+\frac{1}{2},\,y+\frac{1}{2},\,z\right]$	[9]
10	$\left[-y,\,x,\,z+\frac{1}{2}\right]$	[10]
11	$\left[y,\,-x,\,z+\frac{1}{2}\right]$	[11]
12	$\left[x,\,-y,\,\frac{1}{2}-z\right]$	[12]
13	$\left[-x,\,y,\,\frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,z\right]$	[14]

continued ...

Table 16

No.	position	mapping
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[15]
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[16]